Generative Market Simulation (V0)

A Prediction-Market Testbed for LLM Trading Agents

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Agenda

- The platform — engine, agent turn, tools, matching
- The eval — does the LLM trade rationally?
- Live demo

System Positioning & This Week's Goals

What it is

An event-sourced binary prediction-market engine replicating Polymarket-style microstructure — one engine, three faces: terminal CLI, browser dashboard, Python library.

Goals this stage

- Ship a demo command-line interface an agent can drive
- Test whether an LLM takes reasonable actions through that CLI

The Agent Turn: a Forced Four-Stage Flow

1. reads — pull markets / orderbook / news to gather information
2. commit_view — commit per-market YES probability + a one-line plan (orders before commit are rejected)
3. place_order $\times$N — queue orders for round-end execution
4. finish — one line: “what did I learn this round”

Then — matching (system, round end)

Queued orders are ordered by submission time, then matched on the book in that order; resulting fills are written back onto the queued orders.

All agents decide concurrently against the same frozen snapshot (blind submit).

The Information Environment: News & Signals

Private, heterogeneous, decaying signals

Each round, every agent privately sees a noisy estimate of each market's true probability:

$$s = \mathrm{clip}\big(p^* + \mathcal{N}(0, \sigma_t)\big), \qquad \sigma_t = \sigma\left(1 - \rho\,\tfrac{t-1}{T-1}\right)$$

- Agents differ in σ — information quality is heterogeneous
- σ_t shrinks toward the endgame — signals sharpen near resolution
- The LLM never sees p^*; it must infer from prices + signals

Open question — how to design the final news layer?

- $\sim$ 10 scenario markets; 10–20 news items per “day”?
- Public headlines vs. private signals — which carries the edge?

Agent CLI: What Tools Can It Call?

Read tools (6) (free, frozen snapshot)	
get_markets	prices, volume, rounds to resolve
get_orderbook	full bid/ask ladder + depth
get_trade_history	recent public trade tape
get_portfolio	cash, positions, open orders
get_news	headlines; private-signal entry
get_news_detail	full text + reliability

Action / control (4)	
commit_view	YES beliefs + plan
place_order	queue a limit buy / sell
cancel_order	cancel a resting order
finish	end turn + reflection

Not yet built (3) (not_supported, V1)

create_account	fixed at init
create_market	agent-listed markets
transfer	inter-agent transfers

Enforced order each turn: reads → commit_view → place_order/cancel_order → finish.

Matching: One Book, Four Settlement Types

One shared order book

All orders live in a single book in YES-price coordinates (1–99¢). A NO order is mirrored in as NO@n $\equiv$ YES@(100$-$n), so YES and NO compete in the same book. Best crossing orders trade, price- then time-priority.

Each fill settles by what the two crossing sides intend

buy YES $\bowtie$ sell YES	$\rightarrow$	transfer YES shares
buy NO $\bowtie$ sell NO	$\rightarrow$	transfer NO shares
buy YES $\bowtie$ buy NO	$\rightarrow$	MINT a new YES + NO pair (buyers split 100¢)
sell YES $\bowtie$ sell NO	$\rightarrow$	MERGE the pair (release 100¢)

Integer-cent accounting (taker overpay refunded); cash + shares conserved every step. Runs at each round end on the queued orders.

LLM Rationality Evaluation

8 frozen single-step probes, each judged by a deterministic rule (no LLM-as-judge). Neutral prompt — rules only, true probability hidden.

P1	catch mispricing	✓
P2	take profit	✓
P3	empty-book quoting	✓
P4	complementary arbitrage	✓
P5	don't over-trade	✓
P6	budget constraint	✓
P7	information update	✓
P8	order memory	✓

gemini-3.5-flash

- 8 / 8 probes → GO
- Interface perfect: valid JSON, valid actions, 0 hallucination

It reasons, not just passes

- P3: quotes both sides around fair value to make a market
- P4: buys both legs — "asks sum to 85¢, risk-free arbitrage"

Browser dashboard — order book, price chart, fills tape, P&L,
and a per-agent inspector of each LLM's tool-call trajectory.

→ live walkthrough

Thanks!

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